Project	Eagle P3
CTO / WD	
CDRL	

XORAIL

Communications & Systems Integration

Woojin Wayside Interface Test Plan & Procedure

This document describes a preliminary integration and test plan to integrate the onboard Woojin sub-systems with their paired OCC based sub-systems.

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APPROVALS

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VERSION CONTROL

Version Number	Version Date	Change history
0.1	7/15/2012	Initial version
0.4	7/23/2012	Changes as a result of input from subs, discussion with DTP.
0.5	7/24/2012	Minor changes to add in computer need for PA/VMS test.
0.6	7/26/2012	Changes due to review with Woojin, others on 7/25/2012.

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1. INTRODUCTION

1.1. PURPOSE

The purpose of this document is to describe a proposed preliminary wayside integration test to be performed by Woojin. This test shall be supported by Xorail and its subcontractors.

1.2. LIMITATIONS

This document is limited to describing the interfaces and integration aspects of this test. The document does not contain a description of any functional requirements, specifications of the various products used in this test or any other information that is not directly related to the integration aspects.

1.3. OVERVIEW

Woojin have developed an on-board system that includes the following sub-systems:

- Passenger Information Display sub-system.
- Public Address sub-system.
- Communication Control sub-system.
- On-board Wireless LAN sub-system.
- Wayside Network sub-system.
- Passenger Emergency Intercomm.

These various onboard sub-systems have already been integrated and tested from an onboard perspective. Many of these sub-systems provide input to or are interfaced to their respective sub-systems in the OCC (head-end). Woojin would now like to perform an integration test that transcends the on-board side by having the various onboard sub-systems send and receive information from the wayside head end system. This requires the temporary installation of these head-end subsystems in a lab environment.

1.4. DEFINITIONS AND ACRONYMS

-

2. TEST ITEMS

Woojin to provide a list of onboard subsystems that are under test here.

3. FEATURES TO BE TESTED

Woojin to provide a list of those features that are under test here e.g. communication interfaces from Woojin

4. FEATURES NOT TO BE TESTED

Woojin to describe any features that are not to be tested. Since this test is very specific to just onboard-wayside interfaces we can just state that everything else is out of scope.

5. APPROACH

The overall approach for this preliminary integration test plan is to re-create a test environment in the lab that is as close as possible to the actual components being used for the Eagle-P3 project.

Testing will be divided into 2 phases: Internet-based test and Denver-based test. The Internet-based test will be conducted using a VPN established over the Internet with each organization continuing to host their own equipment at their location. The Denver-based test will be conducted in Denver, CO at the Xorail office.

Denver-based Test

Referring to the context diagram below, the intent is to use the actual Woojin components for the PA, PEI sub-systems connected to the HR onboard Ethernet switch. The Nomad CCU will also be connected to this Ethernet switch.

A CCU normally allows for multiple air-interfaces to be provided and manages data transmission from/to these multiple interfaces for train to off-shore communication. For Eagle P3 the intent is to provide cellular (LTE) and WiMax interfaces onboard the locomotive. However, for this preliminary integration plan a LAN connection will be used to simulate a train to offshore bearer. At this time it is not intended to use the RuggedComm Radio for either on-board nor base. The CCU shall be connected to the Home Agent directly.

The OCC end-point of the Nomad CCU system is a Home Agent (HA). This HA will be a demo HA setup hosted by Nomad. Note that the HA will not be co-located with the simulated OCC for this test. So the expected end-to-end RTT's could be relatively high. Ping tests between the CCU, HA and OCC will help in quantifying the delays associated with the remote location of the HA.

Each CCU in each locomotive establishes a connection with the Home Agent. Together these two components are responsible for maintaining the same IP address for the connection as the train traverses the rail network and goes from cell to cell.

For this integration effort the Home Agent shall be connected to a switch. The other components at the simulated OCC are the Penta PA/VMS Simulator and an Avaya switch and phone.

Once the system has been connected and configured correctly it shall be possible for announcements generated from the OCC to be heard on the onboard system. It shall also be possible for messages to be sent from the Penta OCC Simulator for display onboard. Similarly PEI calls made from the onboard system shall be heard in the OCC.

For purposes of interfacing the Penta PA/VMS Simulator to the onboard system it is expected that a train is treated the same as a station i.e. the Penta PA/VMS system will use the same protocol to interface to a train that it uses to interface to the stations.

In order to ensure the correct priority of network traffic between the onboard/wayside Woojin shall provide priority information for the various onboard data streams. This allows the QoS to be setup correctly on the CCU.

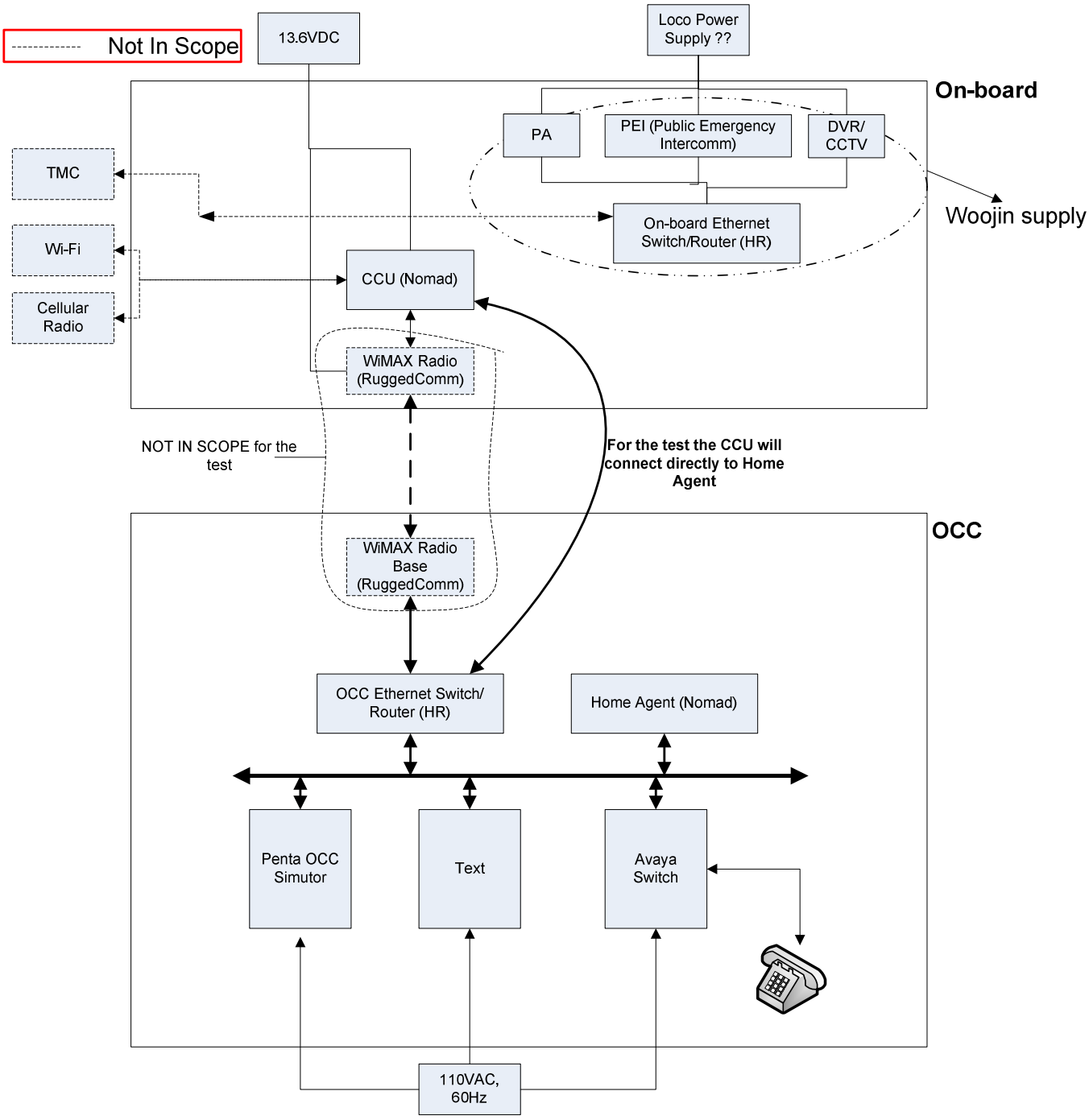


Figure 5-1 Context Diagram

Internet-based Test

In order to maximize the possibility of success when performing the Denver-based integration test it is suggested that prior to conducting this test, integration be conducted via a VPN connection over the internet. To accomplish this one approach is for Woojin to set up a VPN connection that will allow RSC, Penta and Nomad to connect their devices to the VPN connection. This will create a network as described in the context diagram but it will be geographically dispersed.

This approach allows the various sub-systems to be integrated and key interface issues to be debugged. Any major issues found can be researched and fixed and the test repeated. Once most of the system is integrated

and tested then the Denver-based Test can be performed to resolve the remaining configuration oriented issues.

In summary, once the internet based test is complete, testing can also be performed in-person at a lab in the US.

One item that will need to be ensured is that all communication between the onboard and OCC sub-systems is only through the Nomad CCU – HA interface. There must not be any other path that messages can travel through i.e. no loops.

The Internet-based test shall be performed in 2 steps:

Step1: In this step the Nomad devices will not be connected. Instead the Avaya phone sub-system and PA/VMS sub-system will be connected directly to the VPN. Integration shall be performed and any issues found shall be debugged.

Step 2: In this step the 2 Nomad devices shall be add to the VPN network. Data traffic shall flow only between the CCU and Home Agent. The integration tests conducted in Step 1 shall be re-performed. Any issues found shall be debugged. It is expected that any issues found in this step should directly relate to the Nomad devices only.

The figure below shows a VPN established by the VPN Router. The Avaya, Home Agent and PA/VMS Simulator will all be connected to this VPN. Naturally, a local VPN client will be needed, that is co-located at the Avaya, HA and PA/VMS Simulator sites. This local VPN client will be set up to establish the VPN tunnel to the VPN Router in Korea.

TheAvaya, Home Agent and PA/VMS Simulator devices will need to be on one side of the network. The CCU along with the Onboard Ethernet Switch/Router will also be connected to the VPN Router but they will need to be on the other side of the network. The VPN Router will need to keep the 2 networks separate.

The green lines show the network on the OCC side. The Red lines represent the Onboard network. As stated previously the CCU is part of the Onboard network (even though, physically, it will reside in the US).

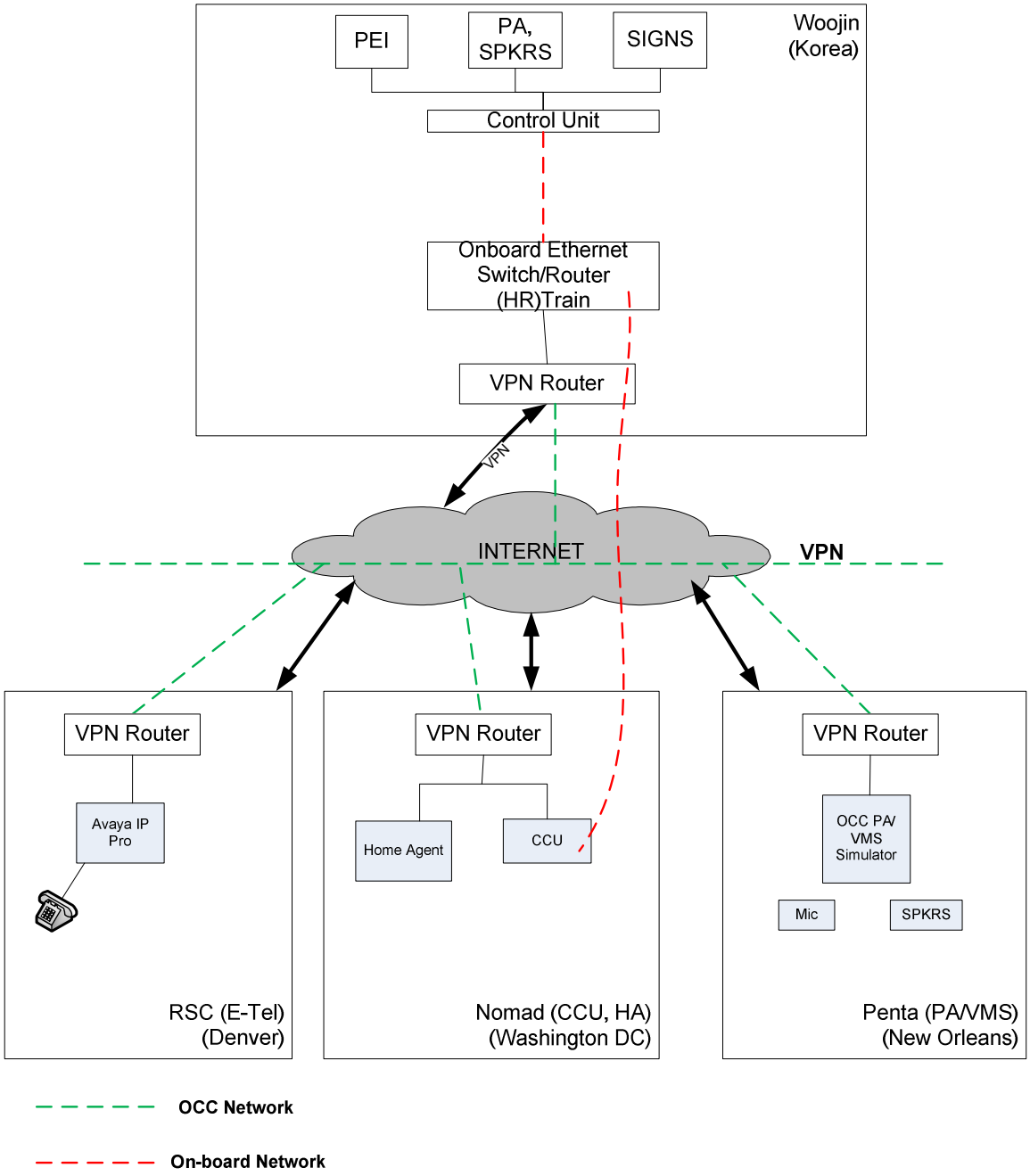


Figure 5-2 Internet-based VPN test

6. ITEM PASS/FAIL REQUIREMENTS

N/A.

7. SUSPENSION CRITERIA AND RESUMPTION REQUIREMENTS

N/A.

8. TEST DELIVERABLES

8.1. TEST DOCUMENTATION

This Test Plan with integrated Test Procedure.

8.2. TEST DATA

This test procedure shall be filled out with test results and submitted to DTP.

9. ENVIRONMENTAL NEEDS

The in-person integration test described in this plan shall require reasonable lab space (at least 2 tables of 6' each with sufficient space around the tables for engineers) to setup all the equipment on benches or tables. It is expected that this test shall be performed in Denver, CO.

Mains power (110VAC, 60Hz) shall be provided.

Each of the participating organizations is responsible for providing their own connecting cables, lab tools, lab equipment, equipment under test, simulators etc.

Bill of Materials

	Item	Quantity	Lead Time
RSC	Avaya IP Pro switch	1	Immediate
	IP Telephone	1	Immediate
	Cables, test equipment, lab equipment, tools	Assorted, to be provided by RSC	Immediate
Penta	OCC PA/VMS Simulator software	1	Depends on compliance to Penta ICD.
	Computer/Laptop for PA/VMS Simulator software	1	Woojin to supply.
	Cables, test equipment, lab equipment, tools	Assorted, to be provided by Penta	Immediate
Nomad	CCU – R3200, Rev C	1	AC version – Immediate DC version – 4weeks
	HA Server (including software installs)	1	Nomad to respond
	FM - Server	1	Nomad to respond
	CCU Firmware, V4.17	1	Immediate

	Fleet Manager, V1106	1	Nomad to respond
	Cables, test equipment, lab equipment, tools	Assorted, to be provided by Nomad	Immediate
Woojin	Train Switch	1	TBD
	PA/VMS simulator computer/laptop (see Penta above)	1	Woojin to supply
	TBD	TBD	TBD

10. RESPONSIBILITIES

Woojin is responsible for providing all the onboard sub-systems that are in their scope. They are also responsible for wiring, configuring all the onboard equipment as well as conducting the overall test. This test is being performed in support of Woojin requirement to conduct a preliminary end-to-end integration test.

Xorail shall be responsible for supporting the in-person test by providing adequate lab space, power and tables/racks to allow all test specific equipment to be wired up. Xorail shall also provide all IP information prior to the test for configuration of the test equipment.

Nomad shall be responsible for providing their equipment, configuring it and supporting the test.

RSC shall provide an AVAYA switch (IP Pro) configured for the onboard IP. Also, an IP phone will be provided and the system will be configured to forward all calls from the onboard to this phone. They are also responsible for configuring the equipment and supporting the test.

Penta shall provide the OCC PA/VMS Simulator and are also responsible for configuring the equipment and supporting the test.

11. STAFFING AND TRAINING NEEDS

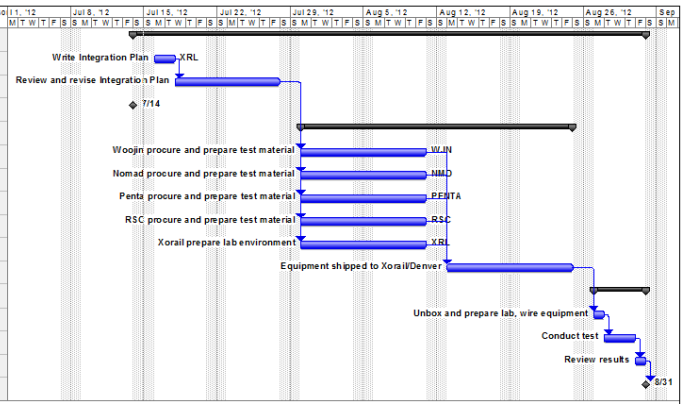
No special staffing needs have been identified for this test other than the obvious responsibility for each company to provide staff in preparation for this test and during the execution of the test.

No special training needs have been identified at this time.

12. SCHEDULE

This schedule below is ONLY a place holder. It is **highly unlikely** that this schedule will be followed. It will be updated after discussions with all involved parties.

ID		Task Name	Duration	Start	Finish	Predecesso
1		**Woojin Prelim Wayside Integration Test**	**35 days**	**Sat 7/14/12**	**Fri 8/31/12**	
2		Write Integration Plan	2 days	Mon 7/16/12	Tue 7/17/12	
3		Review and revise Integration Plan	8 days	Wed 7/18/12	Fri 7/27/12	2
4		Integration Plan released	0 days	Sat 7/14/12	Sat 7/14/12	
5		**Material procurement**	**20 days**	**Mon 7/30/12**	**Fri 8/24/12**	
6		Woojin procure and prepare test material	10 days	Mon 7/30/12	Fri 8/10/12	3
7		Nomad procure and prepare test material	10 days	Mon 7/30/12	Fri 8/10/12	3
8		Penta procure and prepare test material	10 days	Mon 7/30/12	Fri 8/10/12	3
9		RSC procure and prepare test material	10 days	Mon 7/30/12	Fri 8/10/12	3
10		Xorail prepare lab environment	10 days	Mon 7/30/12	Fri 8/10/12	3
11		Equipment shipped to Xorail/Denver	10 days	Mon 8/13/12	Fri 8/24/12	6,7,8,9,10
12		**Perform testing**	**5 days**	**Mon 8/27/12**	**Fri 8/31/12**	
13		Unbox and prepare lab, wire equipment	1 day	Mon 8/27/12	Mon 8/27/12	11
14		Conduct test	3 days	Tue 8/28/12	Thu 8/30/12	13
15		Review results	1 day	Fri 8/31/12	Fri 8/31/12	14
16		Conclude Prelim Integration Test	0 days	Fri 8/31/12	Fri 8/31/12	15



13. RISKS AND CONTINGENCIES

N/A.

14. APPROVALS

Approvals are as listed at the beginning of this document.

15. TEST PROCEDURE

15.1. SPECIAL REQUIREMENTS

List any special tools, software etc required for this test.

RSC has no special requirements at this time.

15.2. SYSTEM CONFIGURATION

What is the configuration and version of all hardware and software ?

i. All devices shall be configured per the IP addresses provided by Xorail.

ii. Onboard: The Woojin system shall be connected to the CCU network. The CCU shall be configured to route all PEI 'calls' via SIP to extension 1001.

iii. Network: The Nomad (simulated) environment shall be configured. The Woojin provided data stream priority for onboard data shall be configured into the CCU (QoS).

iv. OCC: The routing switch from the Nomad network shall be connected to the Avaya IP Pro switch and to the IP phone.

1. The IP onboard shall be configured in the Avaya switch as follows
   a. Switch will accept calls from onboard.
   b. Car ID is programmed into Avaya switch associated with this extension.
2. The OCC phone shall be configured as extension '1001'.

15.3. PROCEDURE STEPS

15.3.1. LOG

Record the results of the tests in the boxes provided on this form.

15.3.2. SETUP

Enter the names of the persons performing the test:

Name	Date of Test	Signature

Describe other setup that has to be performed before the test can start e.g. any configuration parameters that have to be loaded into various equipment, any calibration that has to be done before testing can start etc.

15.3.3. START

Short list of items to be done before this test procedure can be executed.

Nomad: CCU loaded with firmware and configuration file along with correct IP configuration. Pings between the CCU and subsystems should be functional. A laptop connected to the CCU should be able to communicate with the HA. Basic ping/connectivity between laptop and OCC should work. Verify that the above connections work for all the bearers which are being used for the test: WiMax/LAN/LTE. Verify that the CCU is reporting to the FM server.

15.3.4. PROCEED

Step	Test Description	Expected Results	Results
	PEI TEST		
1	PEI initiates call to driver	Rings in cab	
2	Driver does NOT answer within 30 seconds. Call to OCC extension 1001 is initiated	Extension 1001 rings and shows Car ID	
3	OCC phone is answered by tester.		
4	Onboard PEI shall talk	Listening at OCC, rate the voice quality as acceptable	
5	OCC tester should talk	Listening at PEI, rate the voice quality as acceptable	
6	Driver should attempt to answer original call	?????	
7	OCC should hang up	?????	
8	Repeat test steps 1 – 3, and have PEI hang up???	?????	

	PA/VMS TEST (assumes Train is treated the same as a station)		
3.00	**HEARTBEAT MESSAGE From Penta Server** a. Verify that the Penta Server/Simulator sends 'heartbeat' message to Train VLU. b. Verify Train VLU responds to Penta server 'heartbeat' message.	a. Penta active server/simulator sends appropriate heartbet message at designated interval to each active Train VLU. b. Each asctive train VLU responds appropriately top Penta server/simulator 'heartbeat' message.	
3.01	**AD HOC MESSAGE FROM CONSOLE** a. From the console/Simulator, connect to Test Train PA and make an ad hoc announcement. b. From the console, select a group from the map or PA dialog and make an ad hoc message. The group must contain the Test Train PA. c. Remove Train PA from the group and make an ad hoc announcement.	a. The message can be heard at the selected Train PA system. b. The message can be heard at the selected Train PA. c. The message is not heard at the Train PA.	
3.02	**IMMEDIATE MESSAGE** a. Record a new prerecorded audio file named "Test Message." Preview the message using the "Preview" function.	a. The audio message is clear and complete.	

	b. Add "This is a Test" for the text and save the message. Schedule the message for the Test Train PA for 1 minute. c. Edit the message and change the text to "FAT Test." Schedule the message for the Test Train PA for 1 minute.	b. The audio is heard and the text message is displayed at the Test Train PA for 1+/-0.25 minutes. c. The audio is heard and the new text message is displayed at the Test Train PA for 1+/- 0.25 minutes.	
3.03	**SCHEDULED MESSAGE** a. Schedule the prerecorded message "Test Message Priority 2" for priority = 2 and play forever. b. Schedule the prerecorded message "Test Message Priority 3" for priority = 3 and play for 1 minute to begin at 5 minutes in the future.	a. The message "Test Message Priority 2" is displayed on the Test Train PA LED display. b. At the appropriate time +/- 0.25 minutes, "Test Message Priority 3" plays for 1+/-0.25 minutes and then "Test Message Priority 2" is displayed.	

15.3.5. MEASURE

Describe any special measurements that have to be taken during the test, other than those described in the Proceed section.

15.3.6. SHUTDOWN

Describe any special shutdown requirements i.e. does anything special have to be done after the test is complete before the system can be shutdown or the next test started.

15.3.7. RESTART

Describe any special requirements that have to be met before the test can be restarted.

15.3.8. STOP

Describe any special requirements that have to be met before the test can be stopped.

15.3.9. WRAP UP

Are there any special requirements that have to be met before the test can be wrapped-up

15.3.10.CONTINGENCIES

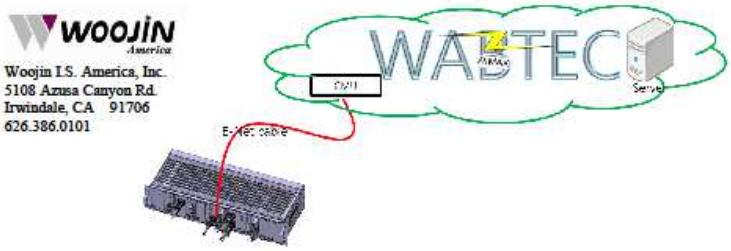
Any special contingencies during or after testing.

15.4. SUMMARY OF TEST RESULTS

Provide a short description.

Pass	Fail

APPENDIX A



Preliminary Wayside Transmission Test Scope

1.0 MDS File Transmission Status Check

1.1 Transmission Method and File Structure

FTP (File Transfer Protocol) is used and stored in OCC FTP Server (RFC 959, RFC 1123)

- LIC Data Type : BIN.
- LIC Data Server Directory : RAC FTP Root / LIC Data
- LIC Data File Name : CarNum_yyyymmddhhmmss.BIN
- LIC File Structure : Refer to QT materials. View was performed in GigWerks In SEPTA.
- LIC File Size : Maximum size of SEPTA is 2M. Size is always changed because the last data received is sent whenever received.

1.2 Testing Method

Number	Contents	Pass/Fail
1	Does Woojin's device (CRA) have FTP interface with the Wabtec sever?	
2	Can files be transmitted to the server from CRA?	
3	The file received in the sever and the file transmitted from CRA are the same?	

2.0 VOIP Voice Communication

Communication method between the OCC and intercoms of each vehicle.

2.1 Transmission Method and File Structure

Communication Method	TCP/IP
VoIP Signaling Protocols	SIP - RFC 3261, 3262, 3263, 3264, 2327
Media Processing(Audio Codec)	Voice Coders - G.711, G.723.1, G.729A/B, G.726, GSM FR, iLBC Wideband Coders – G.722, G.722.2 (WB-AMR), G.711.1 Echo Cancelation – G.168-2004 compliant, 64 m/sec tail length
Telephony Signaling	DTMF – TIA464B Programmable Call-Progress Tones
Packetization	RTP/RTCP Packetization – RFC 3550, 3551, 2198 DTMF Relay - RFC 2833

2.2 Testing Method

Number	Contents	Pass/Fail
1	Is voice communication available between CRA and server's device?	
2	Does the server receive the voice data which is transmitted from CRA?	
3	Does CRA receive the voice data which is transmitted from the server?	

DOCUMENT STRUCTURE:

File Description:	Woojin Wayside Interface Test Plan & Procedure
Parent document(s):	None
Integral documents:	None
Offspring documents:	None
References:	None

Template Version 1.5